Abstract

Framework for assessment of hydrothermal plants in Bavaria

Edgar Fabian Espitia Sarmiento[a] Viktoria Pauw[a], Emilio Sanchez-Leon[b], Ishani Banerjee[c], Kai Zosseder[c]
[a]Leibniz Supercomputing Centre (LRZ), Environmental Computing, Garching, Germany
[b] Bavarian Environment Agency, Munich, Germany
[c] Technical University of Munich (TUM), Chair of Hydrogeology, Munich, Germany

Deep hydrothermal reservoirs are a major energy source with low environmental impact and potentially high economic benefit. However, the effective and efficient exploitation of hydrothermal energy is constrained by the complexity of the geological medium and the associated hydrothermal and geomechanical processes. In the Bavarian Molasse Basin, several hydrothermal stations have been successfully deployed, and energy demand is expected to increase. For local authorities, understanding the effect of hydraulic and thermal interactions of groundwater is crucial. The aim of the project BeM-TG[1] is to develop a user-oriented and comprehensive assessment tool for hydrothermal geothermal energy reservoirs in Bavaria. The framework involves two main blocks: the evaluation of the geothermal stations and the inference of geothermal sites. To evaluate the geothermal stations, geological, hydrogeological, geometric, and operational characteristics are collected to build conceptual and numerical models at regional scale using MODFLOW and at local scale using MOOSE-GOLEM and MODFLOW. Once the numerical models are established, a series of simulations are conducted to explore the parametric space of possible realizations, scenarios, and configurations of the geothermal stations, which provide the foundation and the training dataset of the physics-informed neural network model (PINN). In this stage, the behavior of the hydrogeological and geothermal variables is reproduced. Then, a PINN is developed to infer the effects of new geothermal sites on the variation in pore pressure and temperature. Currently, the workflow is defined, and we are testing the regional and local model for water flow and thermal exchange.

1 BeM-TG: Assessment Model for the Use of Deep Geothermal Energy in the Bavarian Part of the Molasse Basin fopr the German: Beurteilungsmodell für die Nutzung der Tiefengeothermie im Bayerischen Teil des Molassebeckens